Semiconductors

Mass action law

$$np = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$$

$$N_c = 2\sqrt{(2\pi kT m_n^*/h^2)^3} \qquad N_v = 2\sqrt{(2\pi kT m_p^*/h^2)^3}$$

Effective densities of states in C & V bands

$$N_{c[Si]} = 1.45 \cdot 10^{16}\ m^{-3} \quad @\ 300K$$

$$E_{g[Si]} = 1.12\ eV$$

Equilibrium

Doped semiconductor

$$\left.\begin{aligned} n &= n_i \exp\left(\frac{E_F - E_i}{kT}\right) \\ p &= n_i \exp\left(\frac{E_i - E_F}{kT}\right) \end{aligned}\right\} \otimes$$

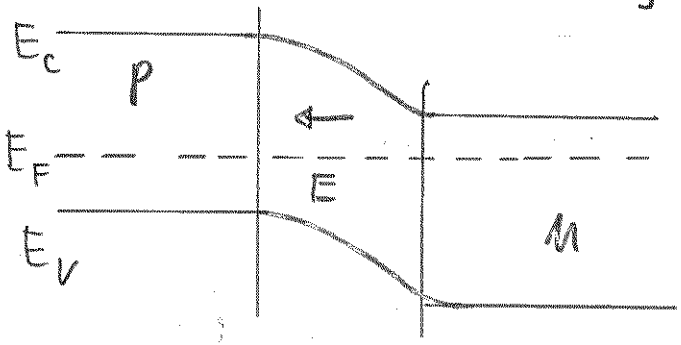
Majority carriers

n-type → $n \simeq N_d$

p-type → $p \simeq N_a$

Minority carriers

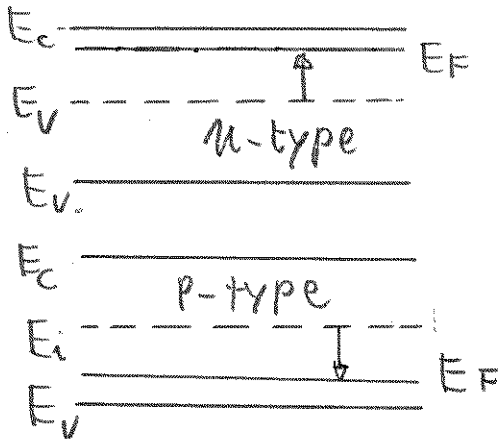
$$\left.\begin{aligned} &n\text{-type} \rightarrow p_n = n_i^2/N_d \\ &p\text{-type} \rightarrow n_p \simeq n_i^2/N_a \end{aligned}\right\} \circledast$$



Intrinsic semicond.

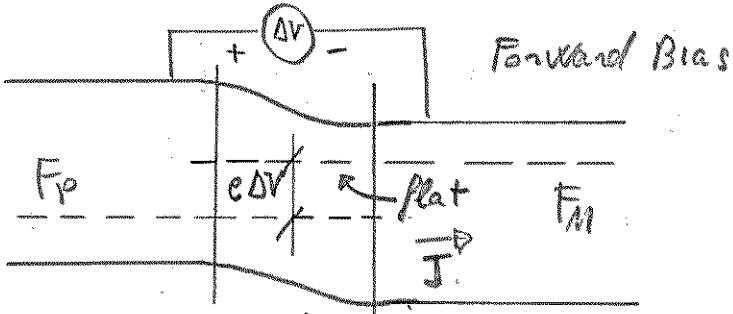
$$n = p = n_i$$

$$E_F \Rightarrow E_i \simeq E_c - E_g/2$$



Stationary state

Biased Junction



$$F_n - F_p = q\Delta V$$

$$\otimes \left\{\begin{aligned} n &= n_i \exp\left(\frac{F_n - E_i}{kT}\right) \\ p &= n_i \exp\left(\frac{E_i - F_p}{kT}\right) \end{aligned}\right\} \text{Majority carriers}$$

$$p_n = n_i^2 \exp\left(\frac{F_n - F_p}{kT}\right) = n_i^2 \exp\left(\frac{q\Delta V}{kT}\right)$$

$$\circledast \left\{\begin{aligned} n_p &= \frac{np}{p} = \frac{n_i^2}{N_a} \exp\left(\frac{q\Delta V}{kT}\right) \\ p_n &= \frac{np}{n} = \frac{n_i^2}{N_d} \exp\left(\frac{q\Delta V}{kT}\right) \end{aligned}\right\} \text{Min carriers edge depl.}$$

Semiconductors

Generation and Recombination

$$R = \alpha\left(np - n_i^2\right) \quad \alpha = \text{constant}$$

P-type semiconductor.

$$R = \alpha\left[\left(n_0 + \Delta n_p\right) p_0 - n_i^2\right]$$ minority carriers.

excess of electrons respect to equilibrium

$$p_0 = N_a \quad n_0 = n_i^2 / N_a$$

n_0, p_0 = carrier densities at equilibrium

$$R = \alpha\left[\cancel{n_0 p_0} - \cancel{n_i^2} + \Delta n_p p_0\right]$$

$$R = \alpha \Delta n_p p_0 \propto \Delta n_p$$

$$R = \frac{\Delta n_p}{\tau_c} \quad \tau_c = \frac{1}{\alpha p_0}$$

With generation rate g @ equilibrium

$$R = g \Rightarrow g = \Delta n_p / \tau_c \Rightarrow \Delta n_p = g \tau_c$$

The excess of minority carriers is proportional to τ_c and g

Current density with Bias

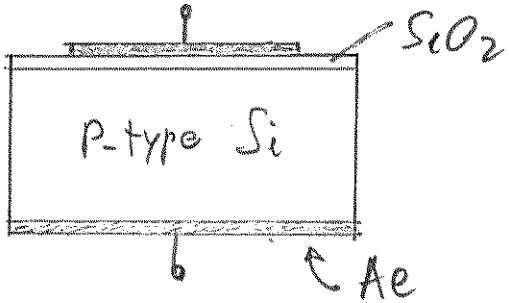
$$\begin{cases} J_n = \mu_e n \dfrac{dF_n}{dx} \\ J_p = \mu_h p \dfrac{dF_p}{dx} \end{cases}$$

With optical generation

$$g_{opt} = R \quad g_{opt} = \Delta n_p / \tau_c$$

$$\Delta n_p = g_{opt} \tau_c$$

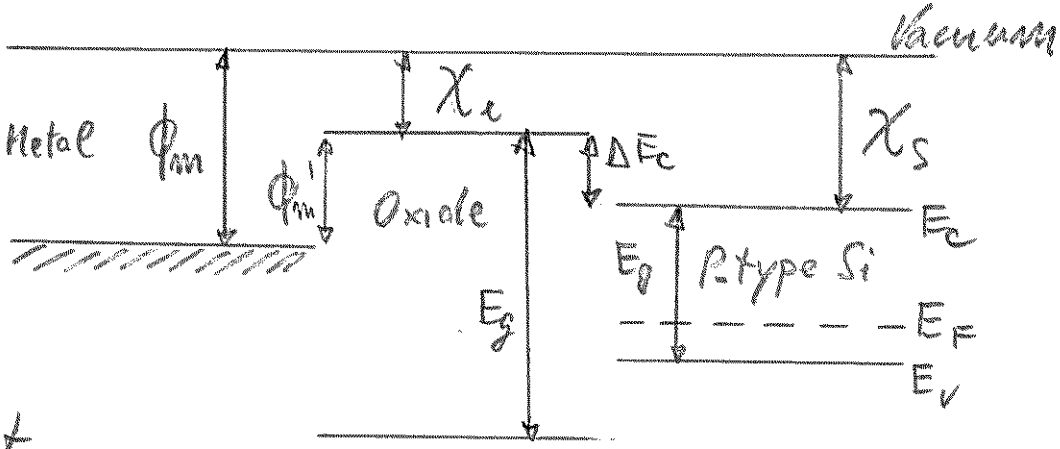
MOS Capacitor



$$\begin{cases} \Phi_{Ae} = 4.1\,eV \\ E_{g[Si]} = 1.1\,eV \\ \chi_{[Si]} = 4\,eV \end{cases} \qquad \begin{cases} \chi_{SiO_2} = 0{,}9\,eV \\ E_{g\,SiO_2} = 8\text{-}9\,eV \end{cases}$$

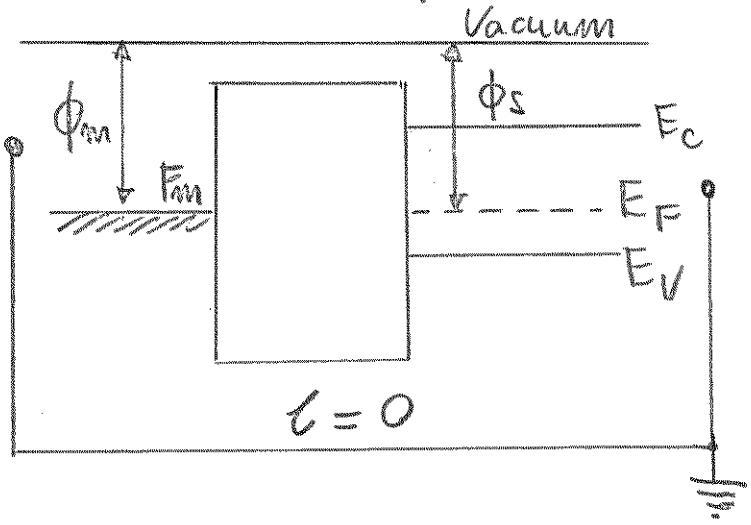
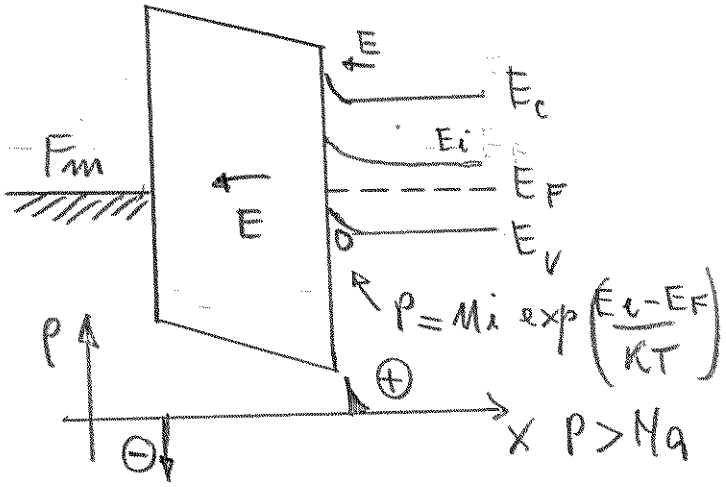
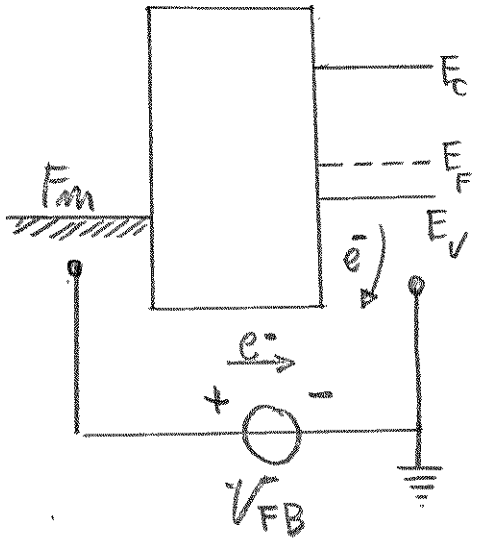
χ = electron affinity

Φ = Work function



$$\begin{cases} E_{c[i]} - E_{c[s]} = \chi_s - \chi_i = \Delta E_c \text{ constant} \\ E_{c[i]} - F_m = \phi_m - \chi_i = \phi_m' \text{ constant} \end{cases}$$

Flat band Hypothesis



$\phi_m > \phi_s$